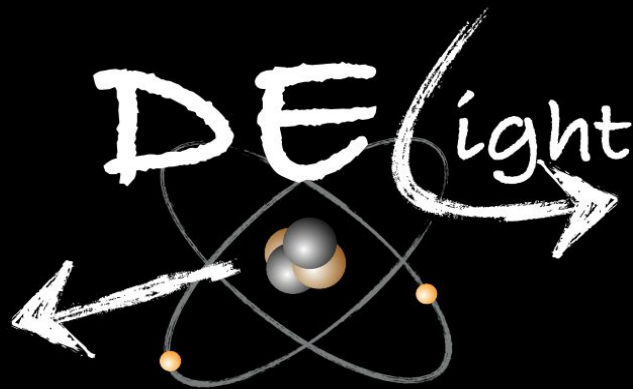
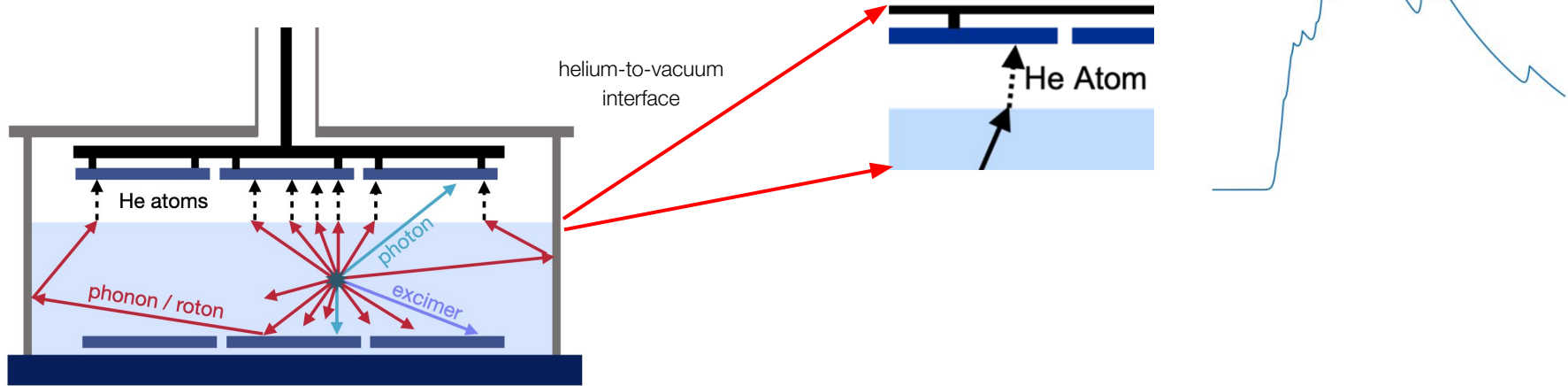


From Detector Likelihood to Geometry-Aware Reconstruction



Physics and signal formation

– Events and physics signature



- **Technical challenge:** Low-energy DELight events produce multi-channel, time-dependent quasiparticle signals. A fixed template is too restrictive because timing, position, channel sharing, and pulse shape vary.

Physical signal model

Assumption for the formation of the signal

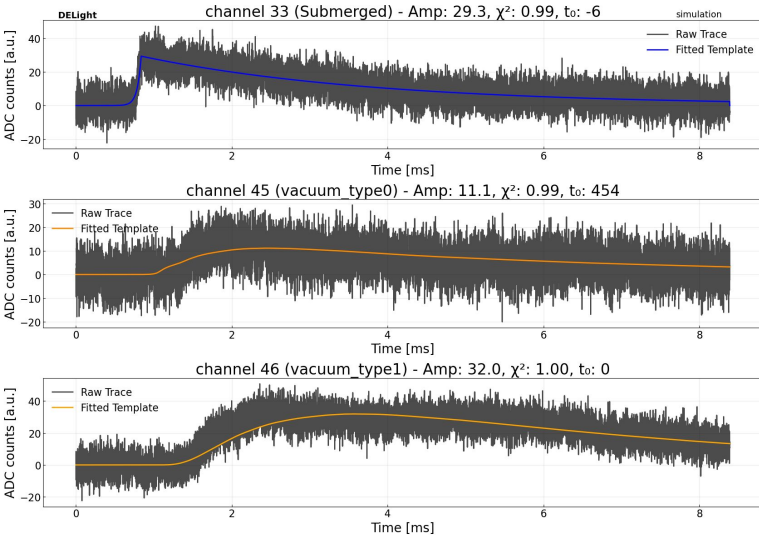
1. Trace = Signal + noise
2. NoisePSD, and assume Gaussian sampling

$$x=s(z)+n, n\sim\mathcal{N}(0,\Sigma), z=(E,t_0,r,\phi,\gamma,\dots)$$

The likelihood determines the loss

Optimal Filter as Constrained Maximum Likelihood

Sample traces fitting for nuclear recoil event @ 500eV

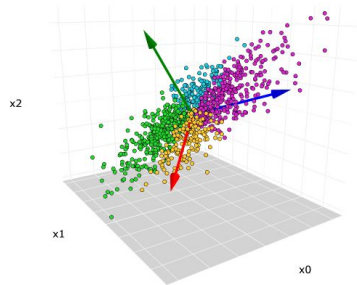


Some formula here

OF is optimal if the signal really lies on one known template direction.

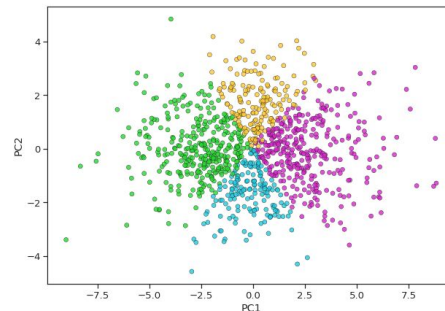
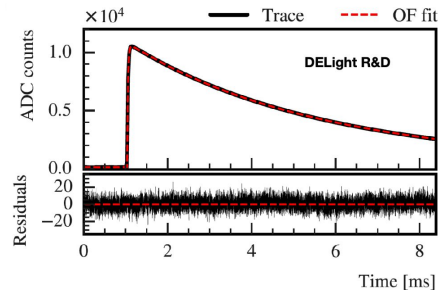
Works well for uv scitilation, but realistic signals vary in timing, geometry, channel sharing, and shape.

PCA: Learning a Signal Subspace



Learn principal signal modes

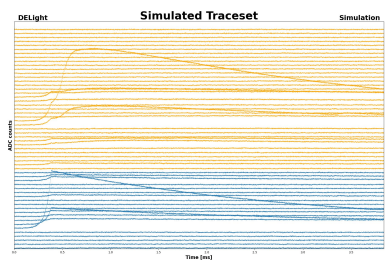
Projection for reconstruction



Instead of fitting traces with a fixed template, PCA **learns the dominant signal subspace** from data.

Signal reconstruction is then performed as a **projection onto principal signal modes**.

EMPCA: Learning Under Detector Geometry

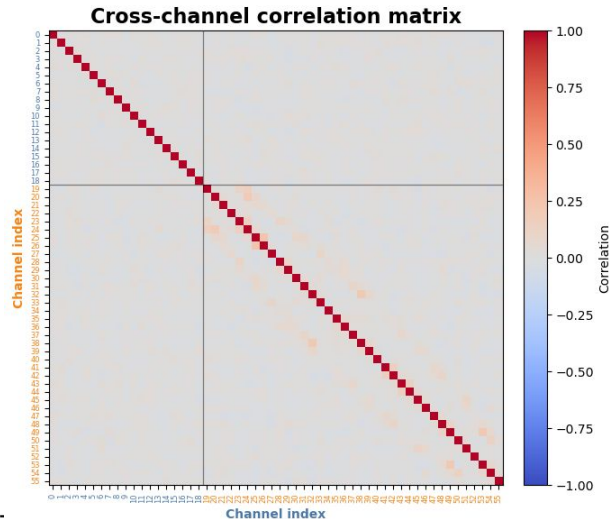


$$\begin{matrix} & \begin{matrix} 1 & 2 & \dots & n \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ \vdots \\ m \end{matrix} & \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \end{matrix}$$

X

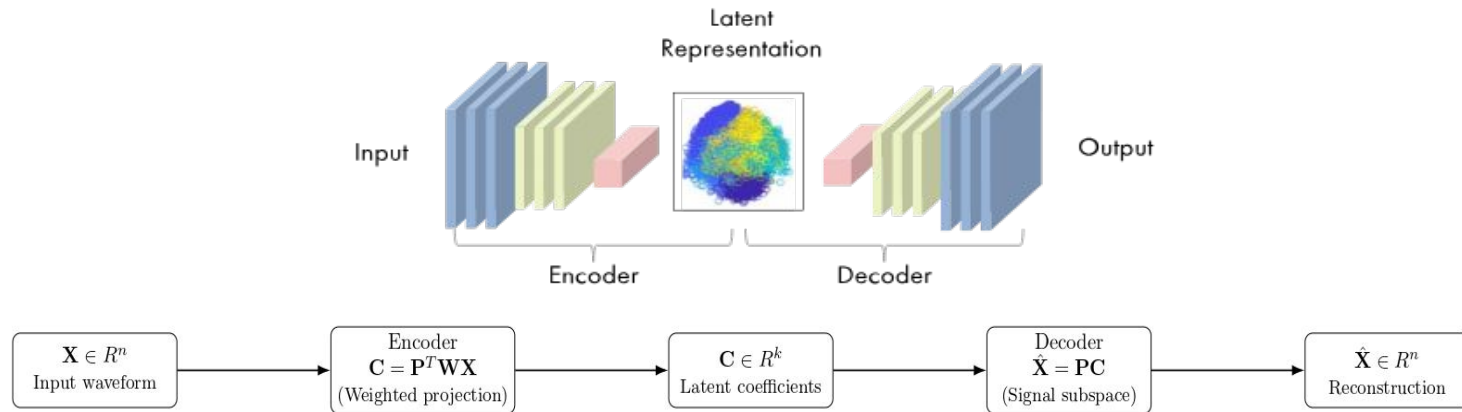
$$\chi^2 = (X - \boxed{PC}) W (X - PC)^\dagger$$

$$\begin{matrix} X \\ \vec{X}_j^{\text{col}} \end{matrix} = \begin{matrix} P \\ P \end{matrix} \begin{matrix} C \\ \vec{C}_j^{\text{col}} \end{matrix}$$



$$\begin{matrix} X \\ \vec{X}_i^{\text{row}} \end{matrix} = \begin{matrix} \vec{P}_k^{\text{col}} \\ P_{ik} \end{matrix} \otimes \begin{matrix} \vec{C}_k^{\text{row}} \\ \vec{C}_k^{\text{row}} \end{matrix}$$

Linear autoencoders inherit the same likelihood geometry



$$c = Wx, \quad \hat{x} = W^T c, \quad \mathcal{L} = (x - \hat{x})^T \Sigma^{-1} (x - \hat{x})$$

EMPCA is mathematically equivalent to a noise-weighted linear autoencoder. Validated by:

- subspace equivalence: principal angle = 0.0046°
- residual equivalence: KS statistic = 9.18×10^{-4}

Under **Gaussian noise**, linear EMPCA provides **optimal linear reconstruction**.

Autoencoders naturally extend this framework to **nonlinear signal manifolds** and **non-Gaussian noise**.

Correct loss geometry is necessary, not sufficient

Mahalanobis loss answers:

How should residuals be weighted under detector noise?

Given a probability distribution Q on $\mathbb{R}^N$, with mean $\vec{\mu} = (\mu_1, \mu_2, \mu_3, \dots, \mu_N)^\top$ and positive semi-definite covariance matrix Σ , the Mahalanobis distance of a point $\vec{x} = (x_1, x_2, x_3, \dots, x_N)^\top$ from Q is ^[6]

$$d_M(\vec{x}, Q) = \sqrt{(\vec{x} - \vec{\mu})^\top \Sigma^{-1} (\vec{x} - \vec{\mu})}.$$

It does not answer:

Which physical factors were present in training?

Which latent directions were never sampled?

Which solution among likelihood-compatible functions should be selected?

NFPA: first structured extension beyond full linear subspaces

NFPA keeps the Mahalanobis likelihood but restricts the reconstruction to a Kronecker channel-time subspace. It is not “EMPCA again”; it is the EMPCA/linear-AE objective restricted to separable channel $\times$ time structure.

Architecture selects among likelihood-compatible solutions

OF: fixed template bias

EMPCA: low-rank subspace bias

NFPA: channel $\times$ time separability bias

CNN/ResNet: locality and deformation stability

Graph model: detector adjacency / channel correlations

Transformer: adaptive long-range token interactions after whitening

Architecture is not a replacement for the likelihood. It is a selection rule inside the likelihood geometry.

Architecture bias: from temporal sequence to channel-time attention

Traditional time-series model:

LSTM / RNN

→ ordered temporal memory

→ good for sequential pulse evolution

→ weaker for explicit multi-channel geometry

Detector signal structure:

time × channel × geometry × noise correlation

Transformer direction:

prewhitened channel-time tokens

→ attention over temporal and cross-channel structure

→ optional noise/precision positional encoding from Σ^{-1}

Architecture is not a replacement for the likelihood; it is an inductive bias inside the whitened geometry.

Wrong loss = wrong geometry (proof-of-necessity)

- PCA objective assumes white noise (Euclidean metric)
- EMPCA objective uses detector metric Σ^{-1}
- Learned subspace, residuals, and reconstruction quality can diverge dramatically

Isotropic PCA: $\min_{P,C} \|X-PC\|_F^2$

Noise-aware EMPCA: $\min_{P,C} \|X-PC\|_{\Sigma^{-1}}^2$

Correct loss is necessary, but not sufficient

- The metric tells you how to weight residuals, not which factors you can learn
- Recovery depends on:
 - coverage: did training data excite the relevant latent factors?
 - inductive bias: does the architecture prefer the right solution among likelihood-compatible reconstructions?

Optional latent vector: $z = (E, t_0, r, \phi, c, \alpha, b, \eta)$

Transformer direction (architecture bias)

- Transformer becomes meaningful when traces are tokenized across channels $\times$ time
- Prewhiten before tokenization to align with likelihood metric: $\tilde{x} = \Sigma^{-1/2} x$
- Optional: noise-aware positional encoding derived from Σ^{-1}

Conclusion

fixed template bias

EMPCA / linear AE

learned low-rank subspace bias

NFPA

channel $\times$ time separability bias

CNN / LSTM

local temporal or sequential bias

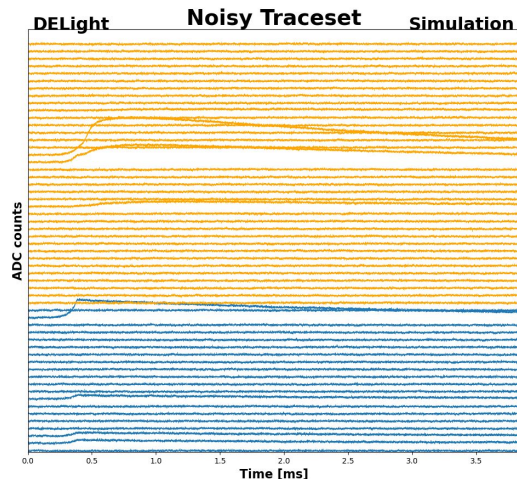
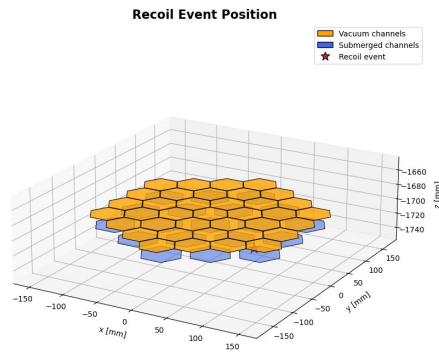
Graph model

detector adjacency / precision-graph bias

Prewhitened Transformer

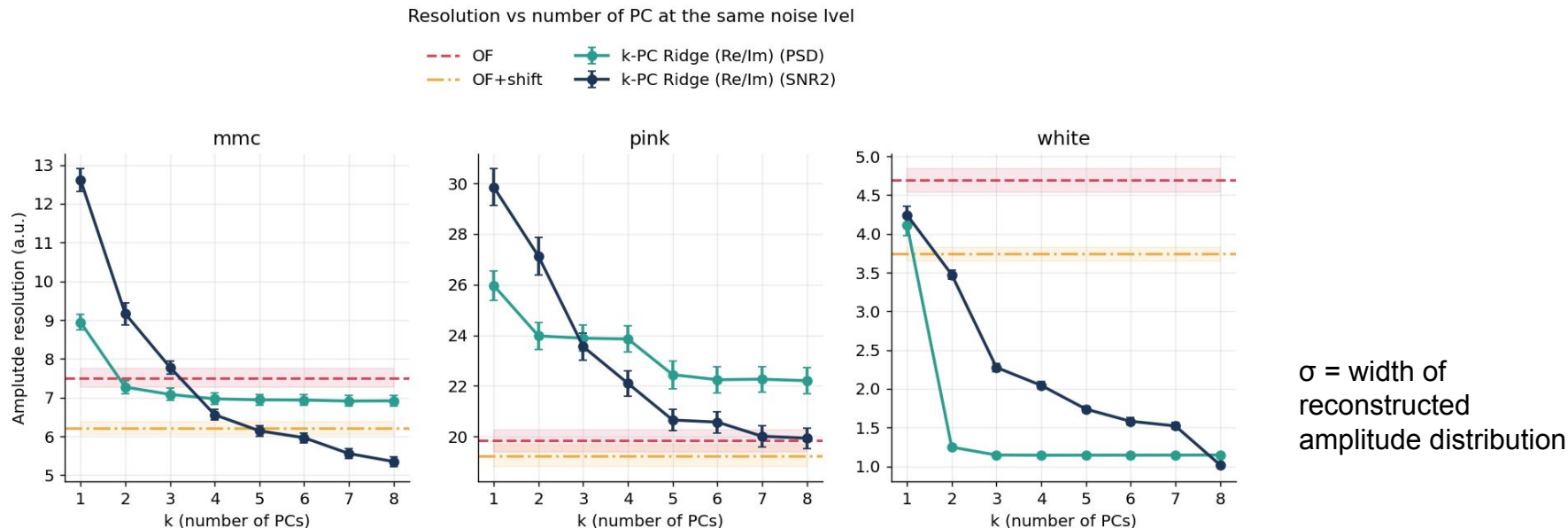
adaptive channel-time attention inside noise geometry

Signal reconstruction in DELight can be formulated as **noise-weighted subspace learning**, where **EMPCA** generalizes the Optimal Filter and yields improved resolution by **extending signal manifold beyond a single template.**



Optimal Filter $\rightarrow$ PCA $\rightarrow$ EMPCA $\rightarrow$ Autoencoder
template linear basis noise-weighted nonlinear manifold

The key result: loss geometry changes what is learned



- EMPCA provides a **learned signal basis**. Amplitude estimated through ridge regression in the latent space.
- This can **outperform the Optimal Filter** when the signal **deviates** from the assumed template.